DATABASE SYSTEMS

Laboratory Work #4

SQL Data Retrieval and Filtering

DBMS	PostgreSQL 18
Client	pgAdmin 4
Database	DB26_Labs
Report language	English

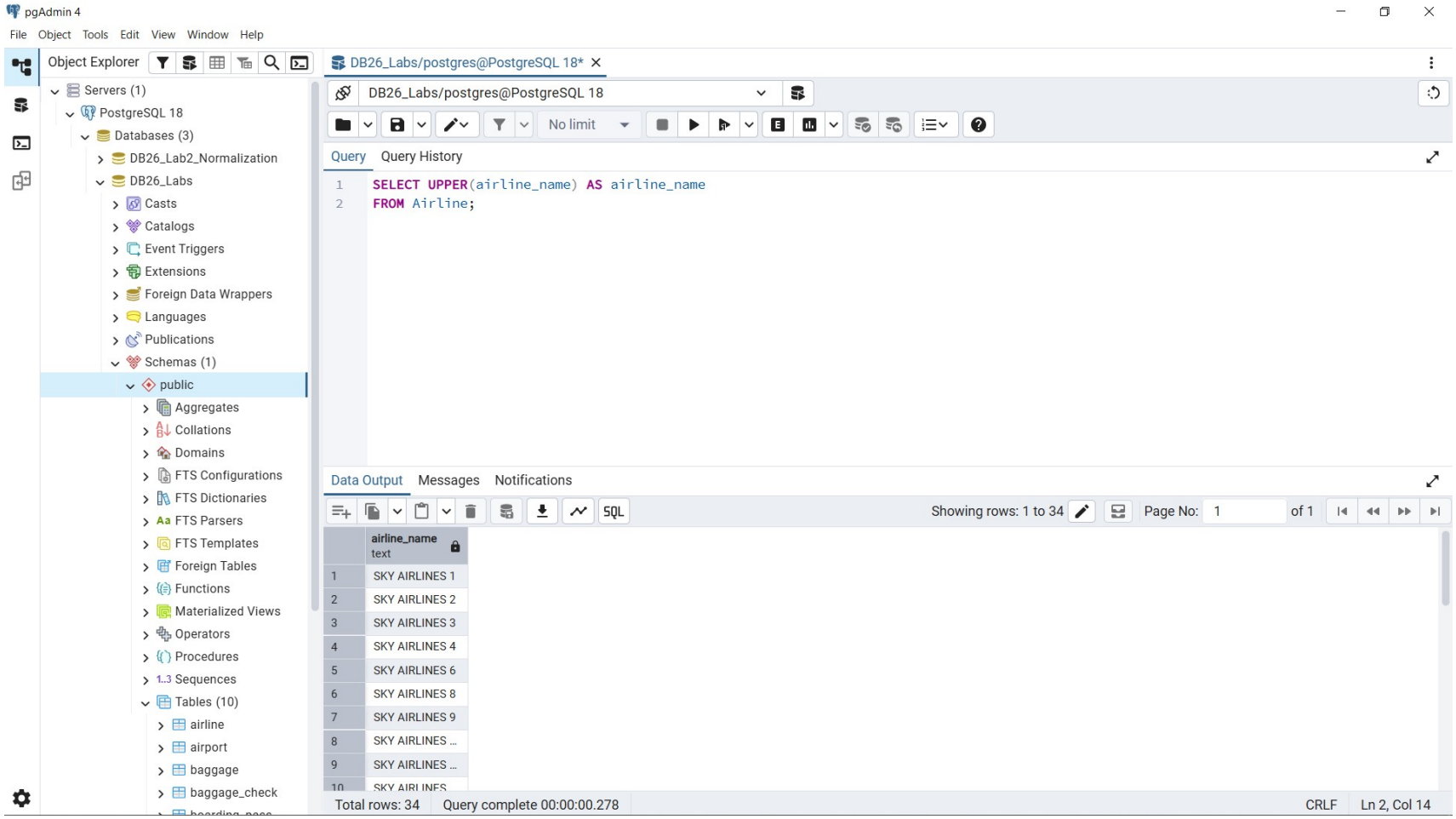
Objective

To practice SQL string functions, date formatting, filtering, sorting, regular expressions, aggregate functions, and result limiting using the airport database created in the previous laboratory works.

Task 1

Retrieve all airline names in uppercase.

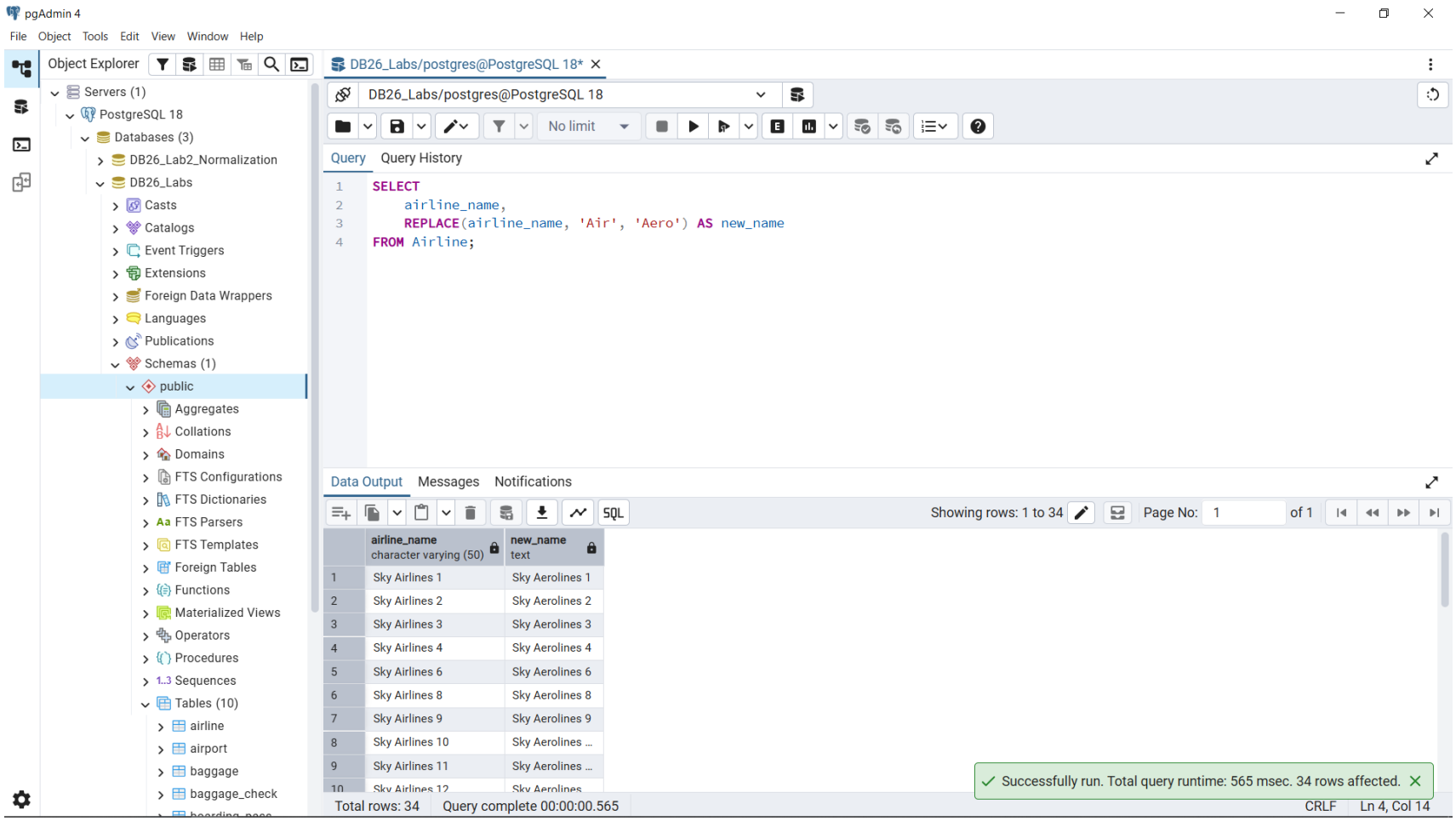
```
SELECT UPPER(airline_name) AS airline_name
FROM Airline;
```



Task 2

Replace every occurrence of the word "Air" in airline names with "Aero".

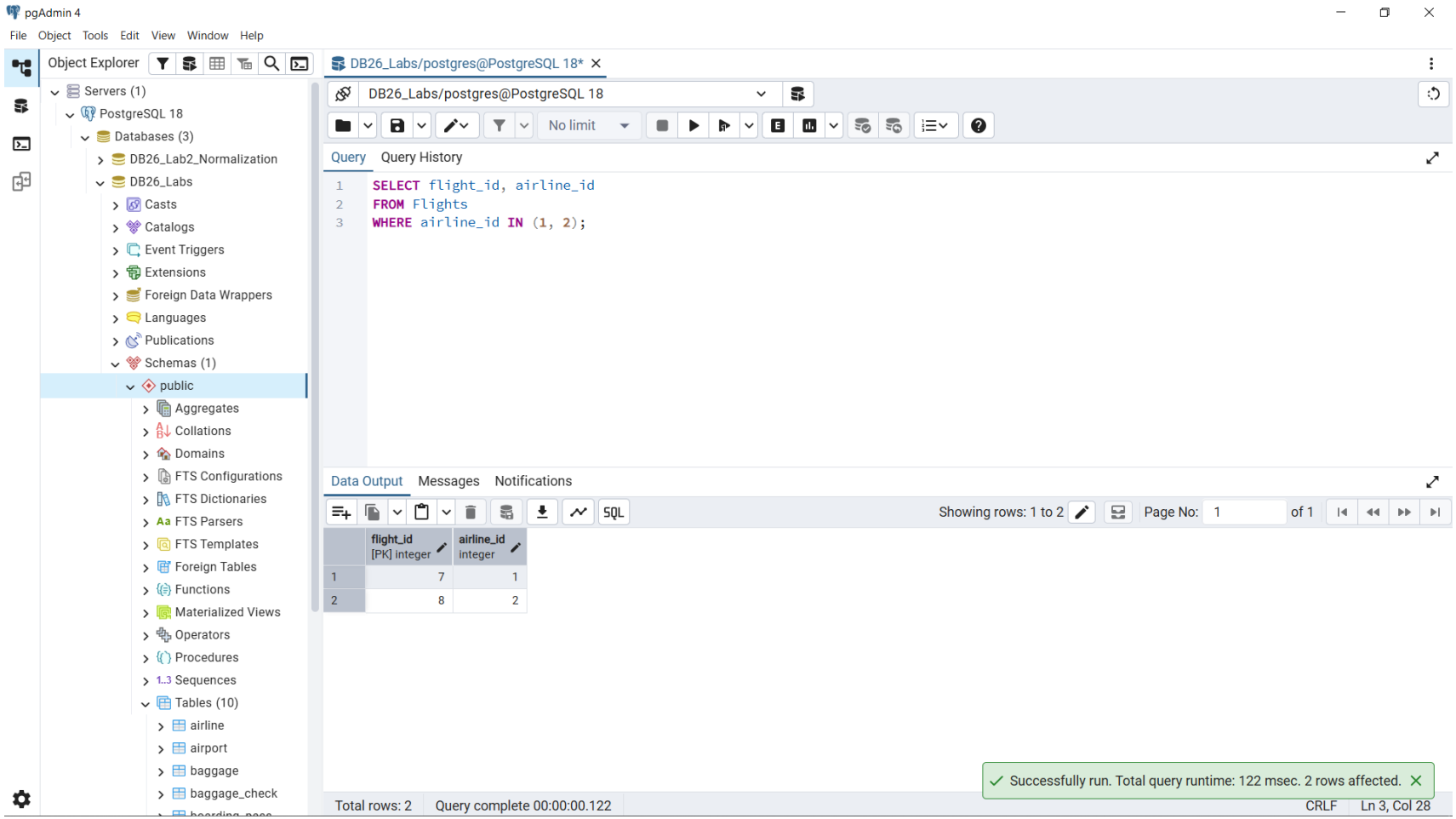
```
SELECT
  airline_name,
  REPLACE(airline_name, 'Air', 'Aero') AS new_name
FROM Airline;
```



Task 3

Find flights related to airline 1 and airline 2. The schema has no separate flight_number field, so flight_id is used as the flight identifier.

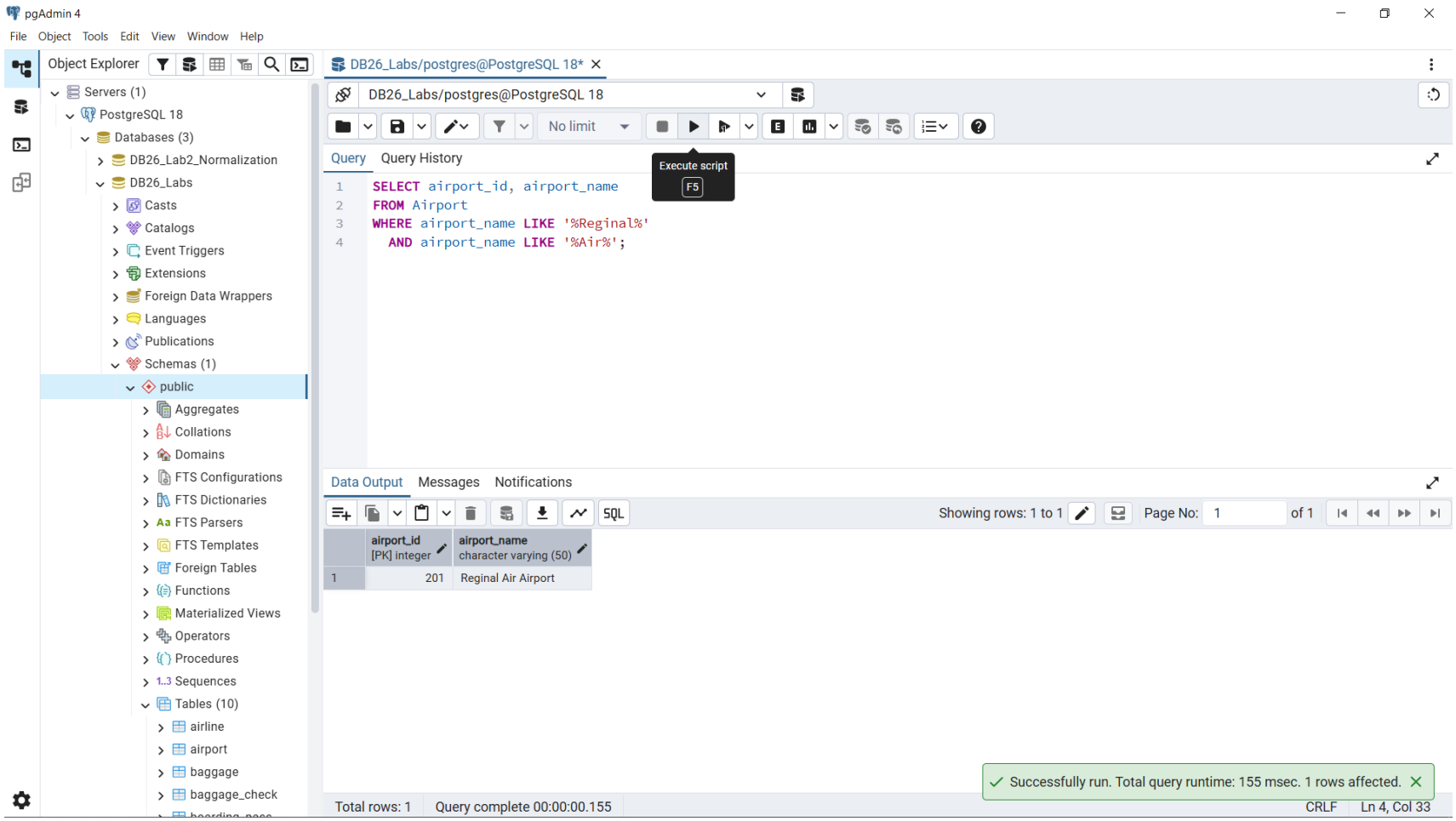
```
SELECT flight_id, airline_id
FROM Flights
WHERE airline_id IN (1, 2);
```



Task 4

Retrieve airports whose names contain both "Reginal" and "Air" (spelling preserved from the assignment).

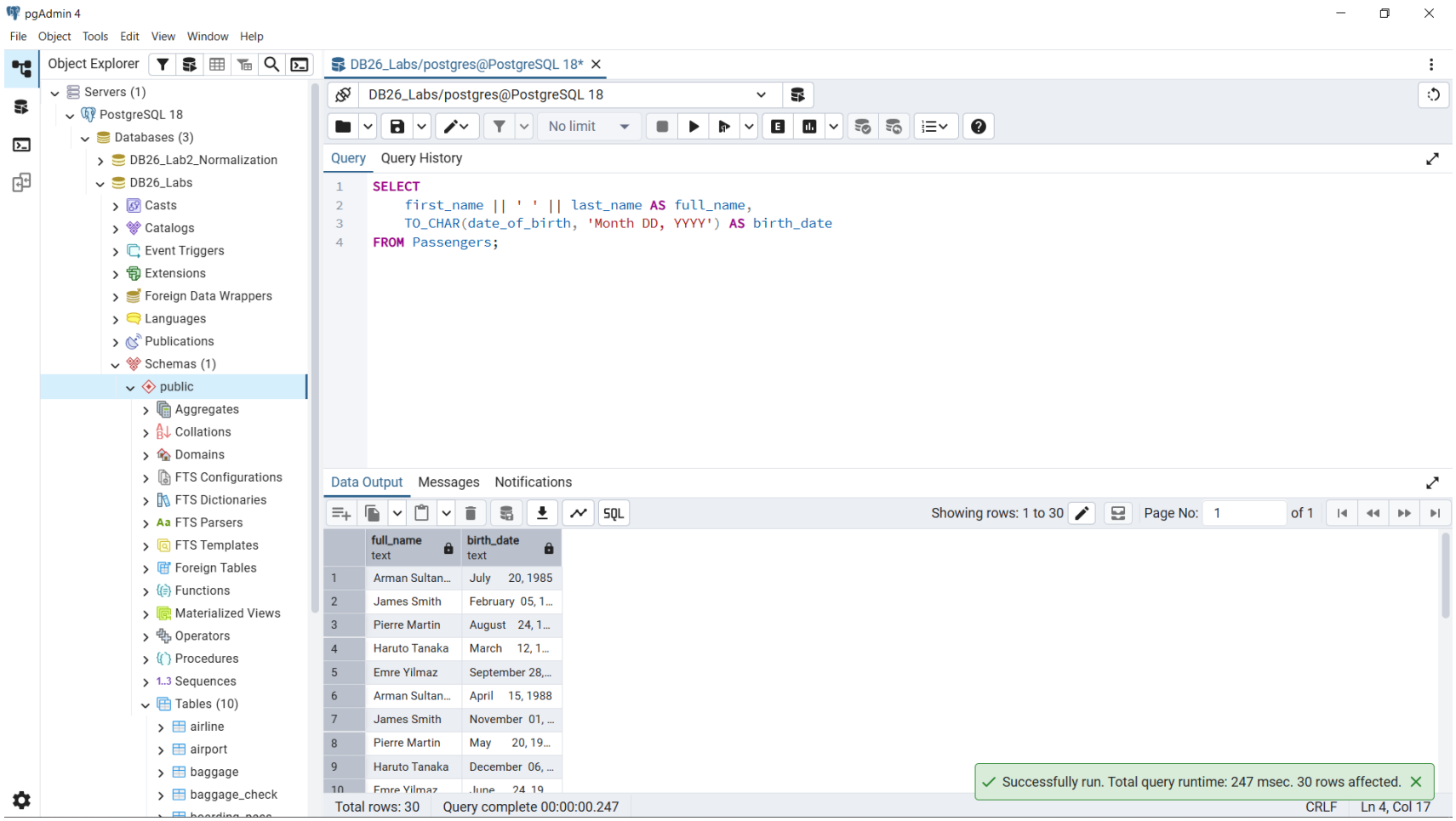
```
SELECT airport_id, airport_name
FROM Airport
WHERE airport_name LIKE '%Reginal%'
  AND airport_name LIKE '%Air%';
```



Task 5

Retrieve passenger full names and format dates of birth as 'Month DD, YYYY'.

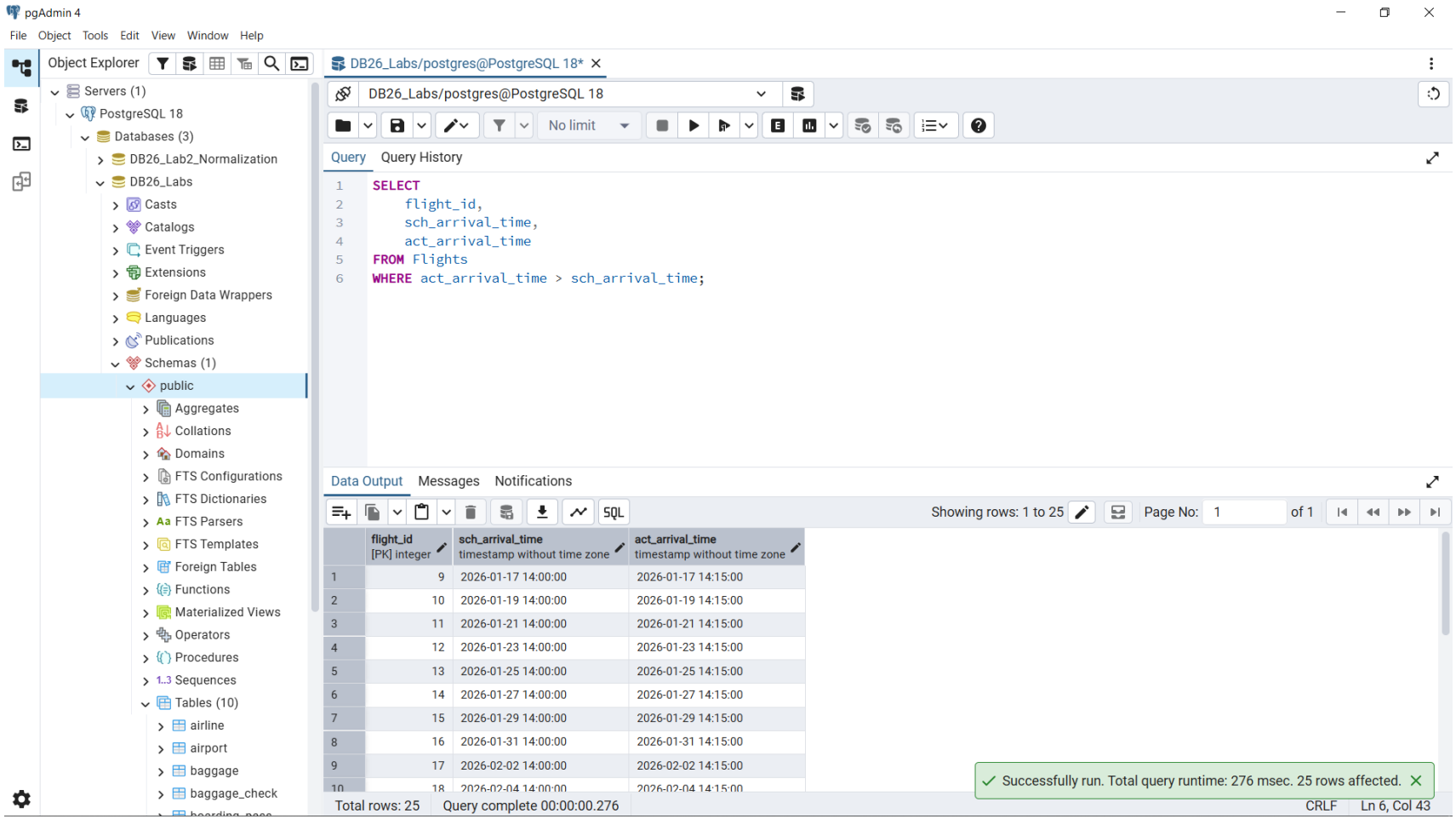
```
SELECT
    first_name || ' ' || last_name AS full_name,
    TO_CHAR(date_of_birth, 'Month DD, YYYY') AS birth_date
FROM Passengers;
```



Task 6

Find flights that were delayed according to the actual arrival time.

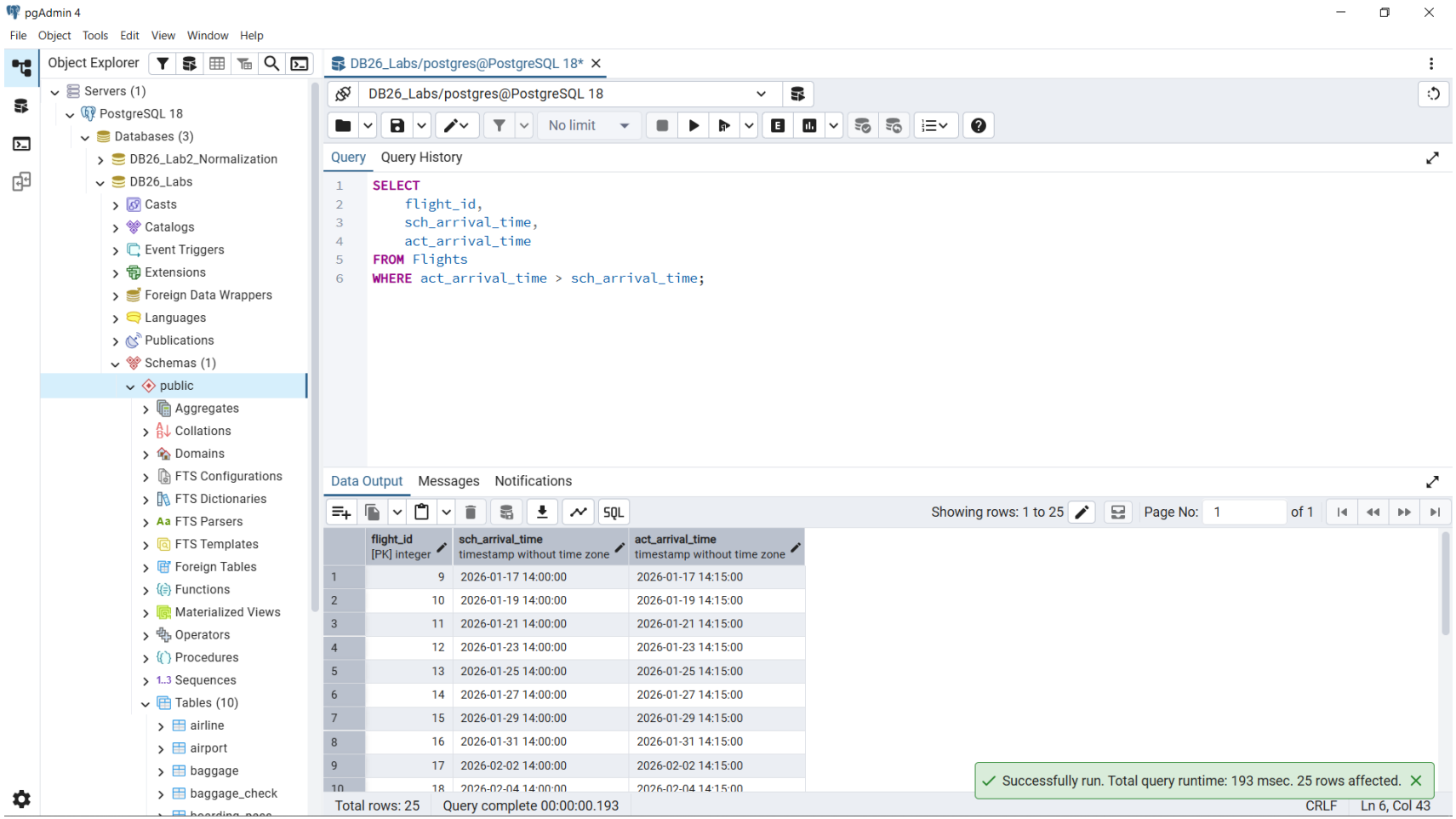
```
SELECT
    flight_id,
    sch_arrival_time,
    act_arrival_time
FROM Flights
WHERE act_arrival_time > sch_arrival_time;
```



Task 7

Find flights that arrived later than their scheduled arrival time.

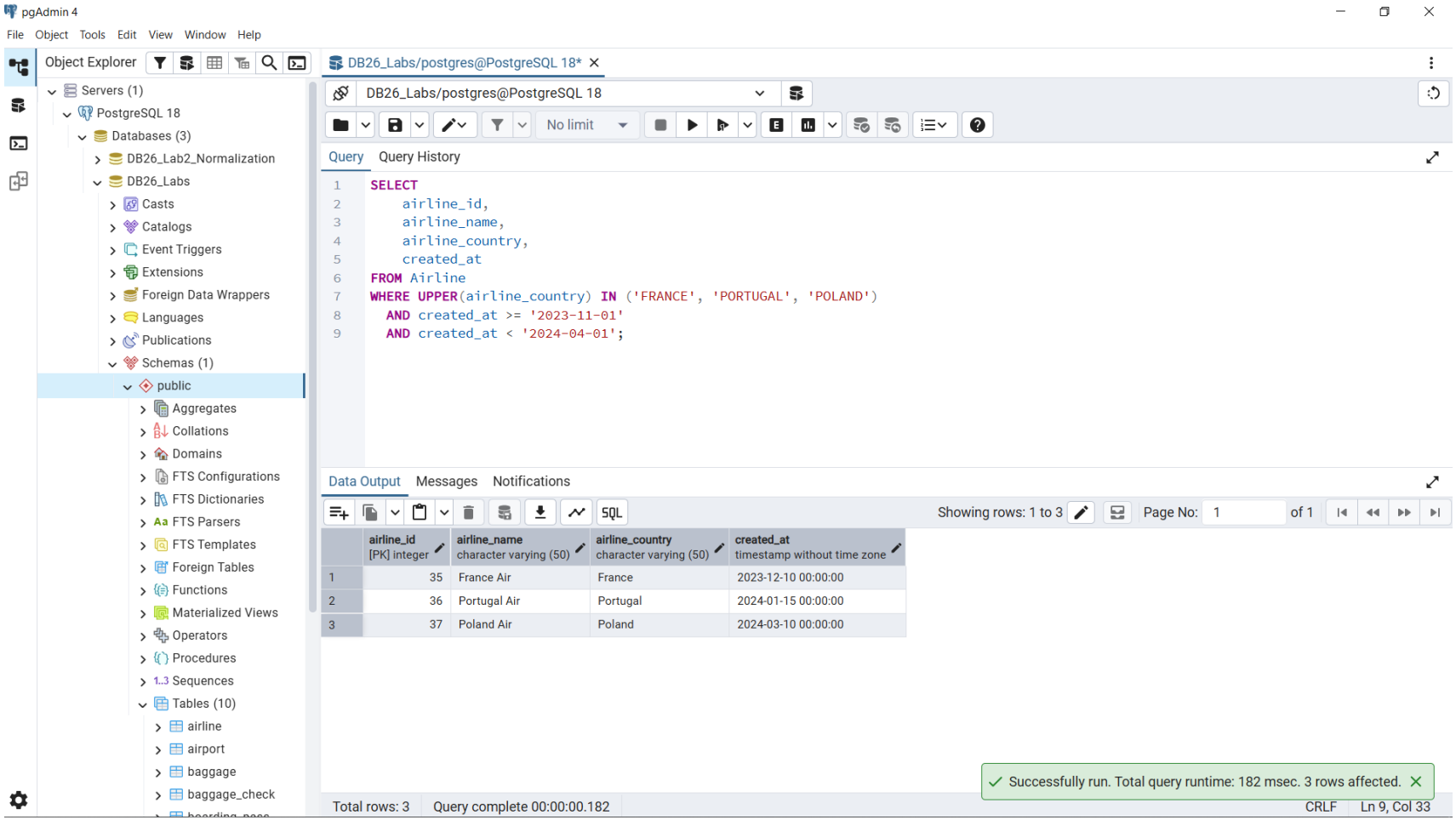
```
SELECT
    flight_id,
    sch_arrival_time,
    act_arrival_time
FROM Flights
WHERE act_arrival_time > sch_arrival_time;
```



Task 8

Show airlines from France, Portugal, or Poland created from 2023-11-01 through 2024-03-31.

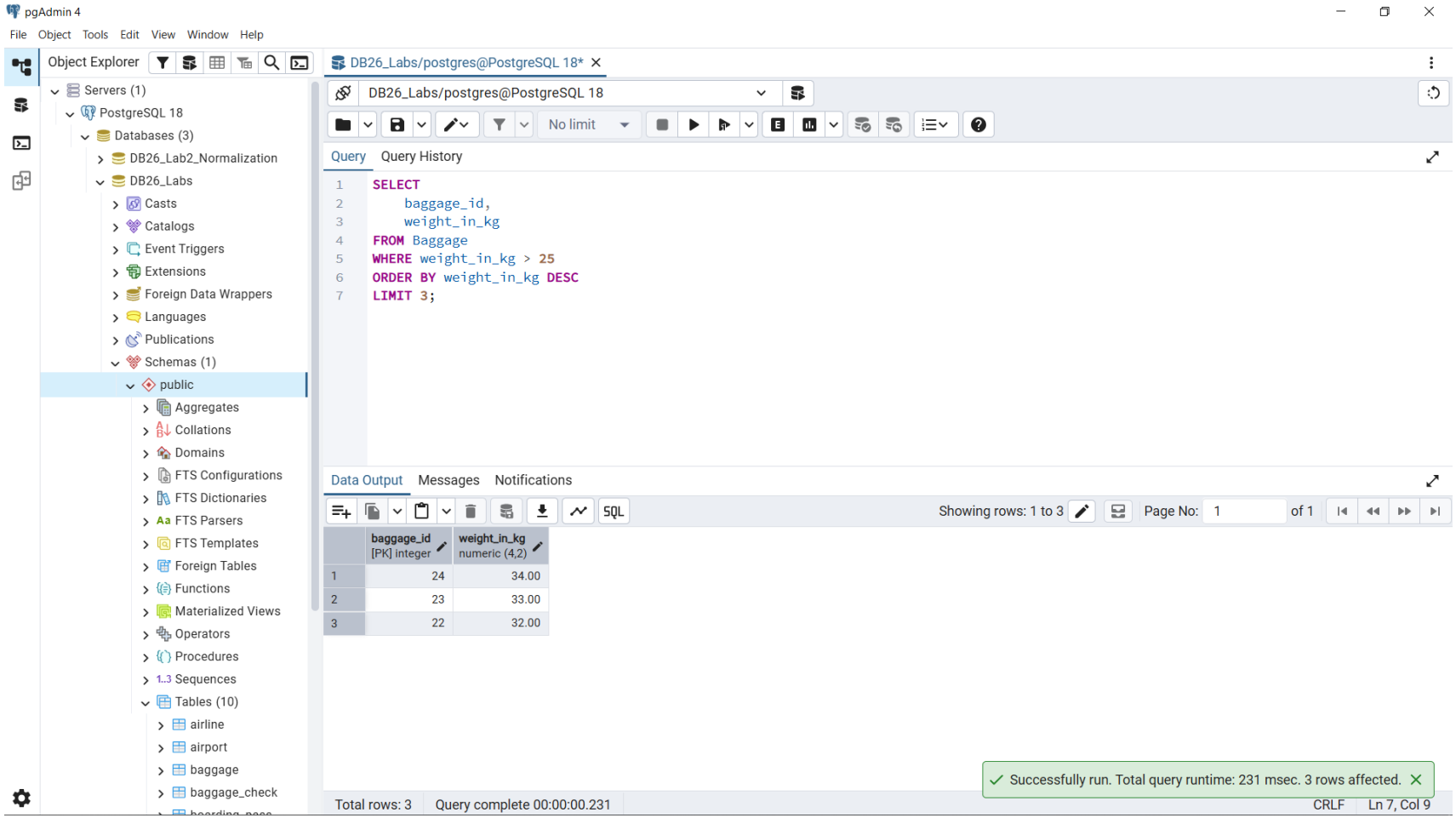
```
SELECT
   airline_id,
   airline_name,
   airline_country,
   created_at
FROM Airline
WHERE UPPER(airline_country) IN ('FRANCE', 'PORTUGAL', 'POLAND')
  AND created_at >= '2023-11-01'
  AND created_at < '2024-04-01';
```



Task 9

Find the top 3 heaviest baggage records with weight greater than 25 kg.

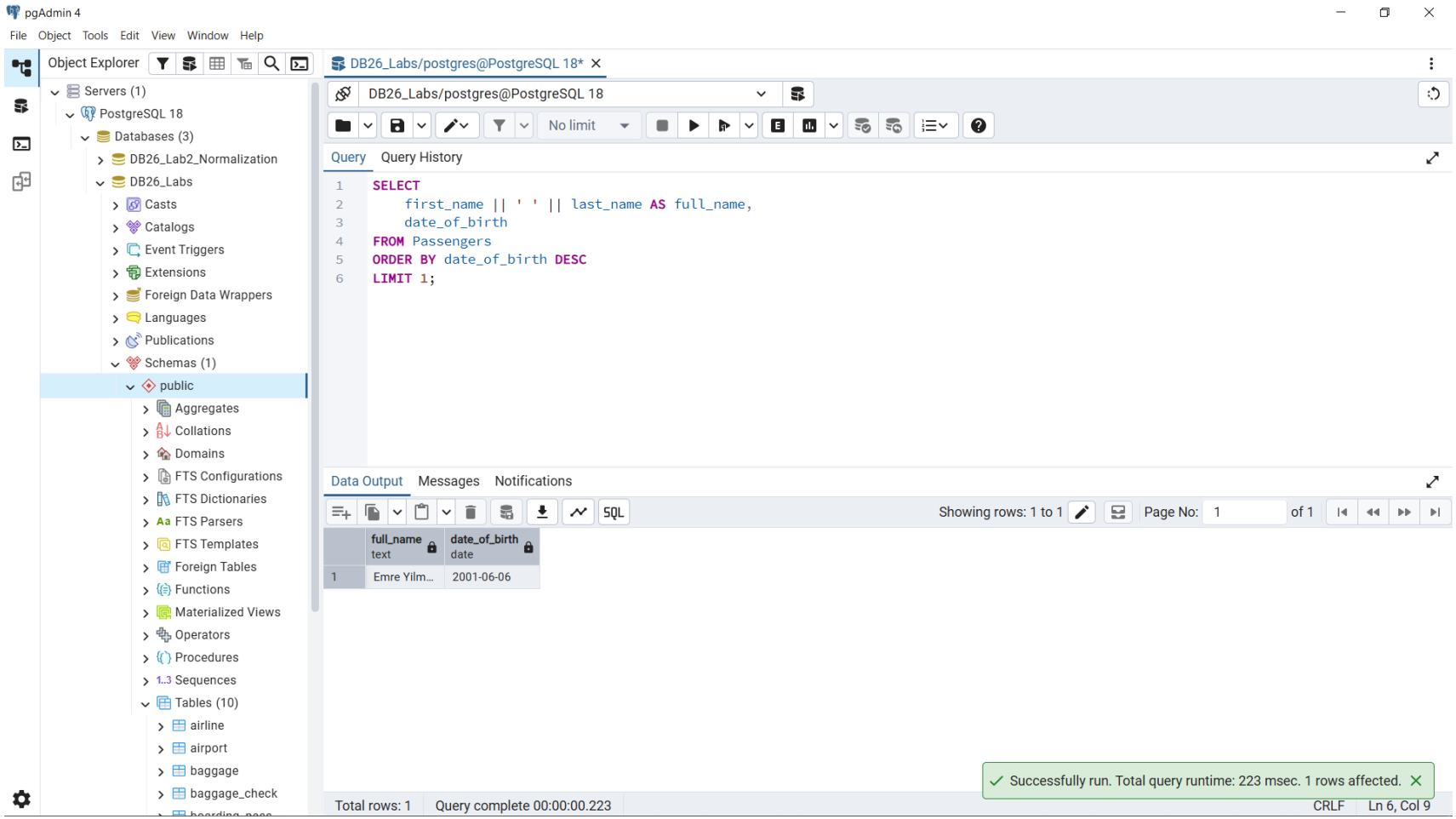
```
SELECT
   baggage_id,
   weight_in_kg
FROM Baggage
WHERE weight_in_kg > 25
ORDER BY weight_in_kg DESC
LIMIT 3;
```



Task 10

Find the youngest passenger's full name.

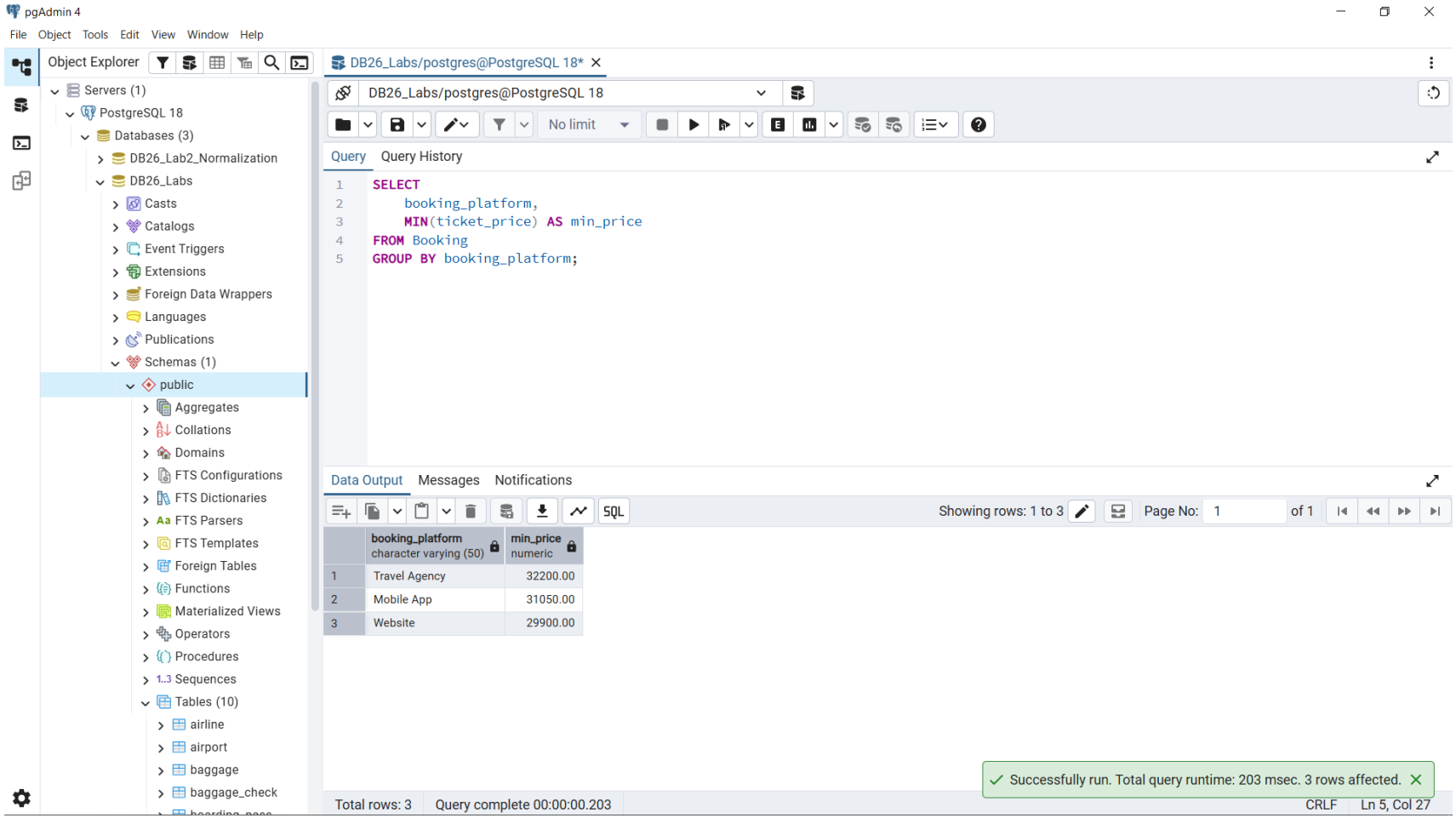
```
SELECT
    first_name || ' ' || last_name AS full_name,
    date_of_birth
FROM Passengers
ORDER BY date_of_birth DESC
LIMIT 1;
```



Task 11

Find the cheapest booking price for each booking platform.

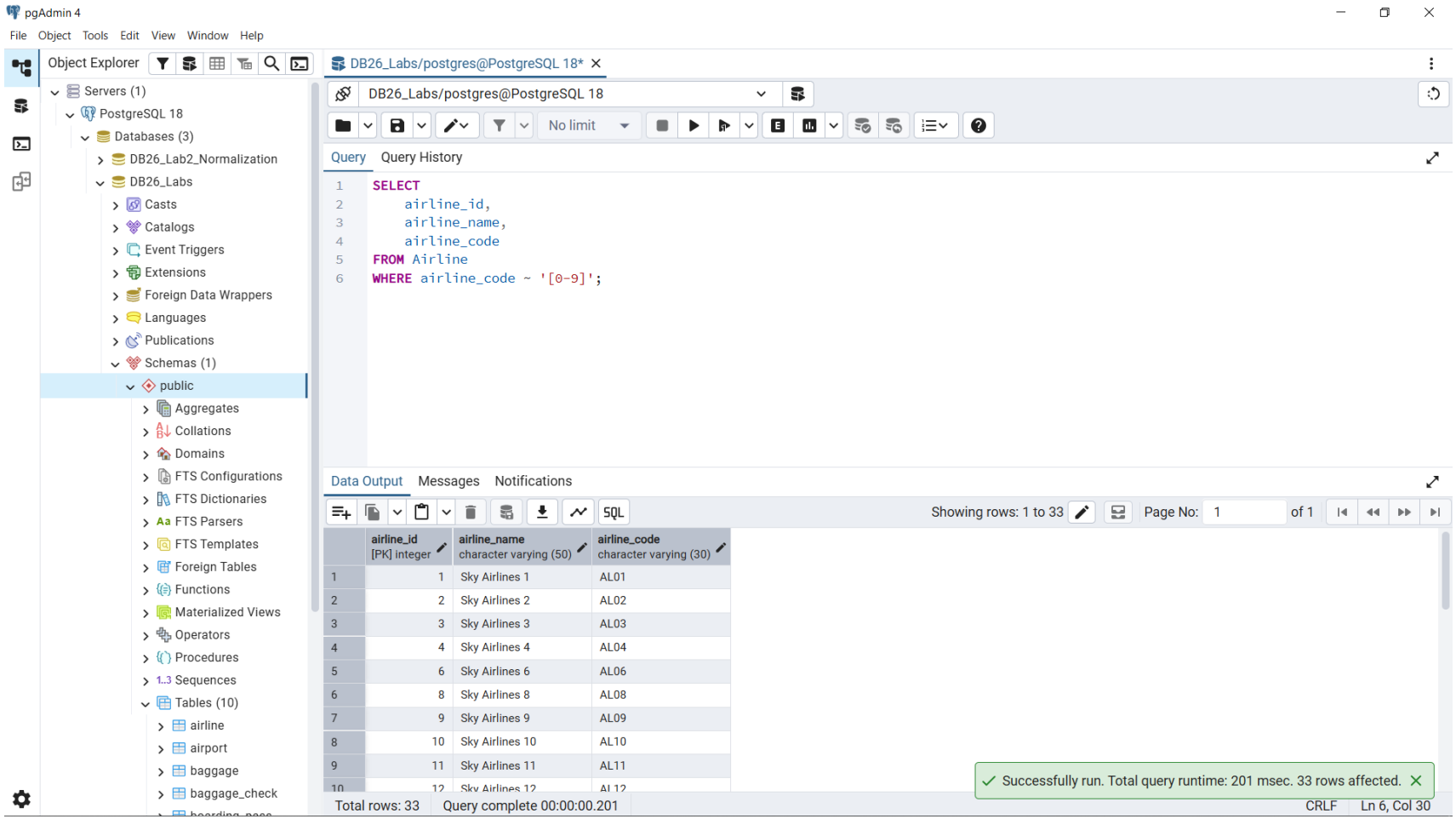
```
SELECT
   booking_platform,
   MIN(ticket_price) AS min_price
FROM Booking
GROUP BY booking_platform;
```



Task 12

Return airlines whose airline_code contains at least one digit.

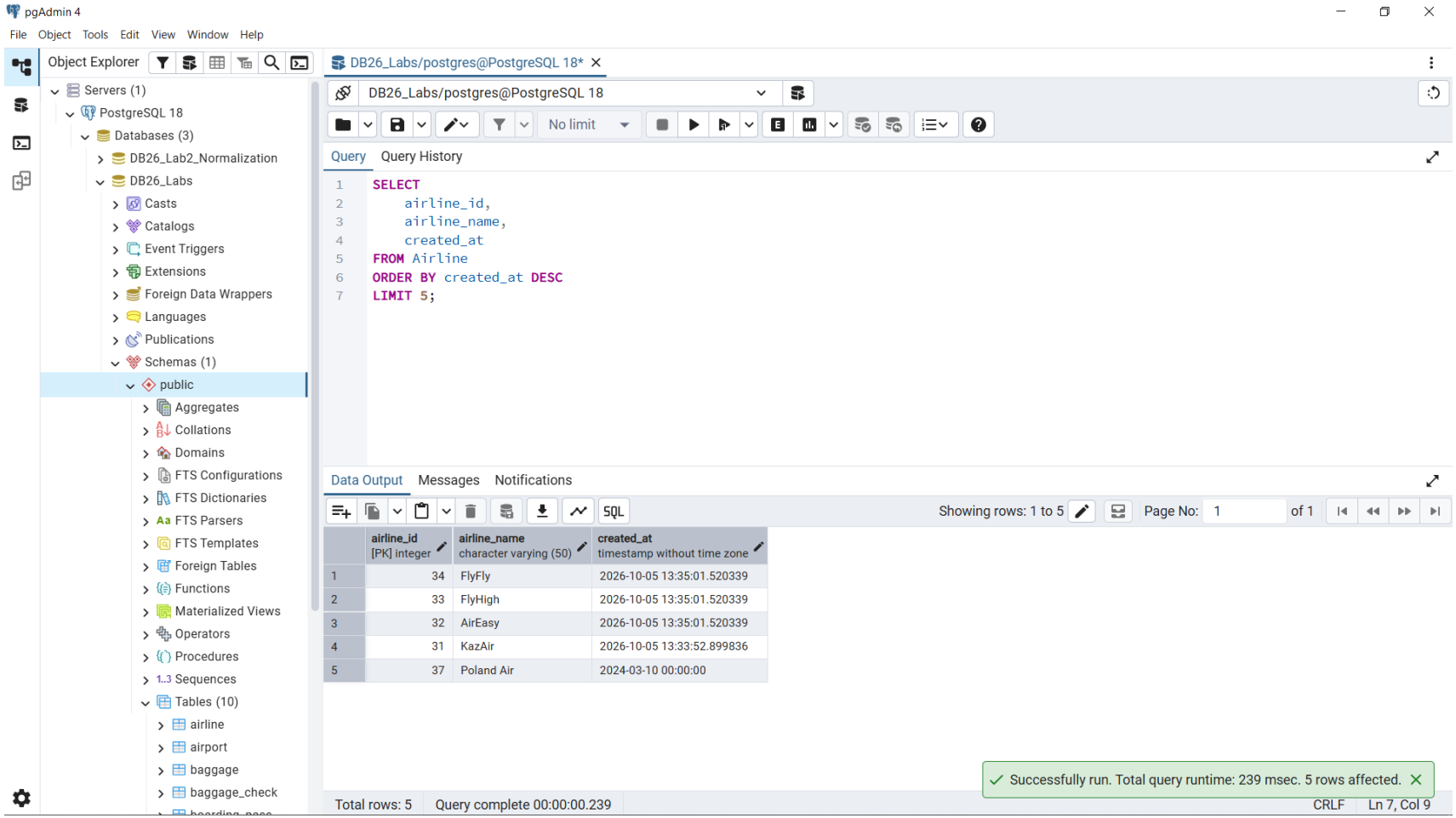
```
SELECT
    airline_id,
    airline_name,
    airline_code
FROM Airline
WHERE airline_code ~ '[0-9]';
```



Task 13

List the 5 most recently created airlines.

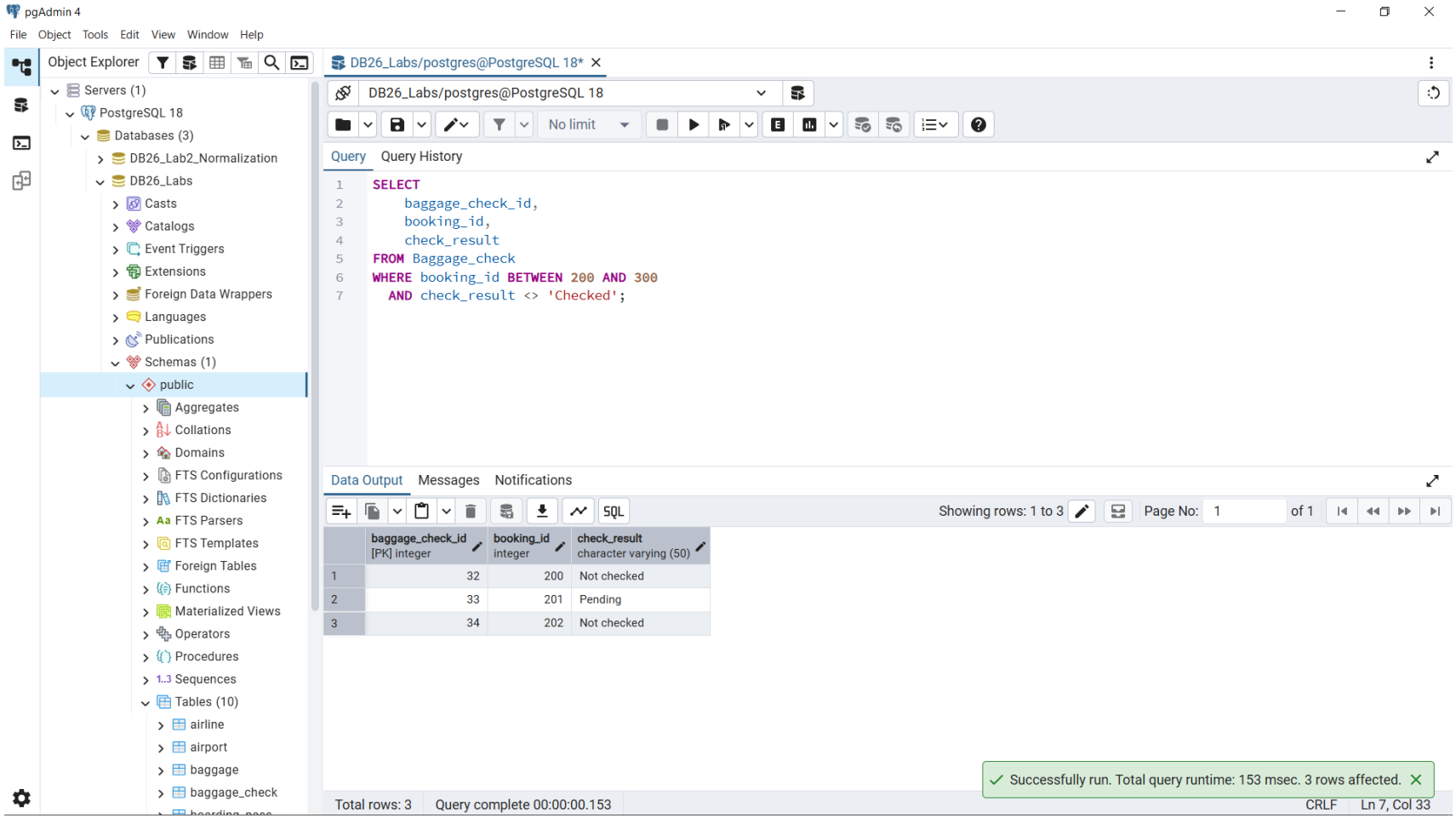
```
SELECT
   airline_id,
   airline_name,
   created_at
FROM Airline
ORDER BY created_at DESC
LIMIT 5;
```



Task 14

Return baggage-check rows where booking_id is between 200 and 300 inclusive and check_result is not 'Checked'.

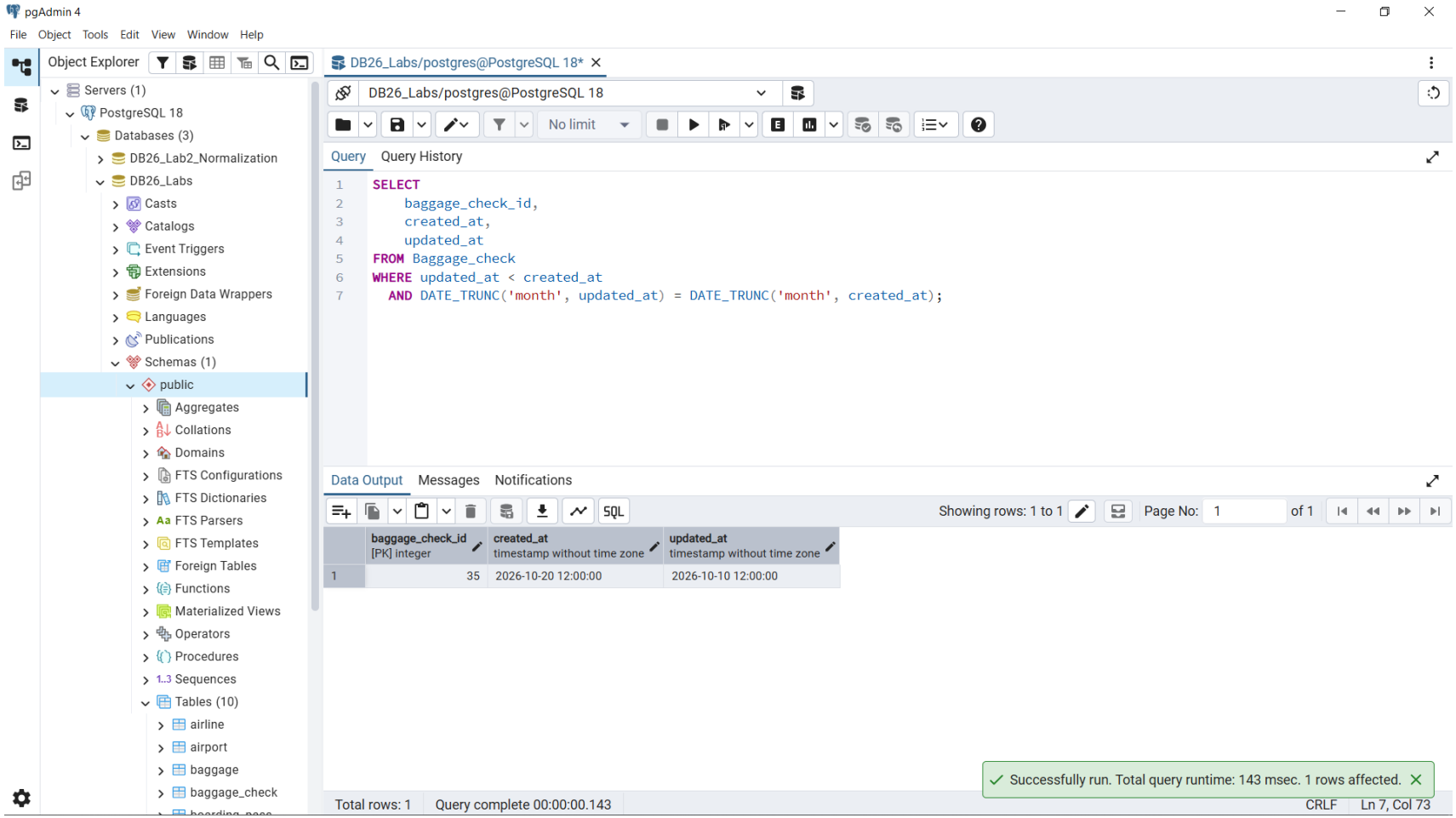
```
SELECT
   baggage_check_id,
   booking_id,
   check_result
FROM Baggage_check
WHERE booking_id BETWEEN 200 AND 300
 AND check_result <> 'Checked';
```



Task 15

Find baggage checks where updated_at is in the same month as created_at but is earlier than created_at.

```
SELECT
   baggage_check_id,
   created_at,
   updated_at
FROM Baggage_check
WHERE updated_at < created_at
 AND DATE_TRUNC('month', updated_at) = DATE_TRUNC('month', created_at);
```



Conclusion

All fifteen tasks were completed in PostgreSQL using the DB26_Labs database. The work covered string transformations with UPPER and REPLACE, pattern matching with LIKE and regular expressions, date formatting and date comparison, filtering with IN and BETWEEN, sorting and LIMIT, and aggregation with MIN and GROUP BY. The screenshots confirm successful execution of each query and display the resulting data in pgAdmin 4.